Modelling and Parameter Estimation of Diesel Engine Aftertreatment System under Catalyst Saturation for Aging Diagnostics

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Abstract: Diesel engine aftertreatment systems, particularly the Selective Catalytic Reduction (SCR) system with Ammonia Slip Catalyst (ASC), are critical for reducing harmful NO_x emissions. The effectiveness of the SCR catalyst degrades over time due to aging, leading to non-compliance with emission standards. The existing approaches for on-board aging detection are often impeded by the complexity of SCR dynamics, the cross-sensitivity of commercial NO_x sensors to ammonia, and the general lack of on-board ammonia sensors in commercial vehicles. This work presents a novel approach for modeling the SCR system and estimating its parameters to diagnose catalyst aging, even with uncertain and limited data from commercial trucks.

We propose a discrete-time, input-output model for NO_x reduction derived from molar conservation across the control volume of SCR-ASC with focus on the dynamics when the catalyst is saturated with ammonia. In this "saturated" mode, the NO_x reduction, $\eta(k) = u_1(k-1) - x_1(k)$ where u_1 is the inlet NO_x concentration at step $(k-1)$ and x_1 is the outlet NO_x concentration at step (k), becomes independent of urea dosing. The NO_x reduction is modeled as an explicit function of inlet NO_x concentration (u_1), the residence time (τ), which is a function of the mass flow rate, and the average concentration of adsorbed ammonia (γ), which strongly depends on the urea-dosing rate. Under catalyst saturation, γ reaches its maximum temperature-dependent capacity, $\Gamma(T)$. Thus, the NO_x reduction in the saturated mode, η_{sat}, is: $\eta_{sat}(k+1) = u_1(k)\tau(k)k_{scr}(k)\Gamma(k)$, where k_{scr} is the SCR rate constant. This saturated model is parameterized using Chebyshev polynomials for temperature dependence, resulting in a linear-in-parameters regression model: $\eta_{sat}(k+1) = \phi_{sat}^T(k)\,\theta_{sat}$. A crucial observation is that the NO_x reduction under saturation, $\eta_{sat}(k)$, forms a tight upper bound for the actual NO_x reduction, $\eta(k)$, at any given time. $\eta(k) \leq \eta_{sat}(k)\ \forall k$. This inequality allows us to estimate the saturated model parameters, θ_{sat}, from operational data where the catalyst's saturation state is unknown.

The proposed methodology is validated using test-cell data, demonstrating consistent parameter estimates across datasets and robustness to sensor cross-sensitivity. Furthermore, we show a clear, statistically-significant difference in parameters between new and aged catalysts using both test-cell and real-world truck data. Using this distinction, a Wald-test-based aging detector is developed. The method is shown to be robust to sensor cross-sensitivity and to provide a reliable metric for detecting catalyst aging, paving the way for more accurate and robust on-board aging diagnostic systems.

Keywords: Diesel Engine Aftertreatment System, System Identification, Switched Nonlinear Systems, Parameter Estimation, Convex Optimization, Statistical Estimation and Detection

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